

SCRIPT-CONTROLLED ACL

phase : planning phase

Date	25 september 2026
Team ID	SWTID-2026-7377
project name	Script controlled ACL
Maximum marks	4

Here is the Project Planning Phase documentation for the Script-Controlled ACL Micro-Project:

Project Planning Phase

1. Work Breakdown Structure (WBS)

To ensure systematic execution within the allocated 3-hour timeframe, the project tasks are divided into four primary work streams:

- Stream 1: Environment & User Setup
 - Elevate privileges to security_admin.
 - Provision target user (EEE User) and assign email credentials (eeeuser@gmail.com).
 - Create custom roles (bb1, bb2, bb3, bb4) and map them to EEE User.
- Stream 2: Data Architecture
 - Create custom table Institution Details (u_institution_details).
 - Configure custom dictionary fields (Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, Description).
 - Populate sample records across multiple choice options (ECE, EEE, CSE).
- Stream 3: Security & ACL Implementation
 - Configure Read ACL with conditional script logic (Branch == EEE & bb1 role check).
 - Configure Create ACL with role bb2.
 - Configure Write ACL with role bb3.

- Configure Delete ACL with role bb4.
- Stream 4: Testing & Verification
 - Perform user impersonation tests (EEE User vs. System Administrator).
 - Verify CRUD operations (Create, Read, Update, Delete).

2. Timeline & Schedule (3-Hour Milestone Plan)

Time Frame	Milestone Task	Focus Area & Deliverables
0:00 - 0:30	Milestone 1: User & Role Setup	Create EEE User and create/assign roles bb1, bb2, bb3, bb4.
0:30 - 1:15	Milestone 2: Data Model Configuration	Build table u_institution_details, configure fields, and insert sample data.
1:15 - 2:15	Milestone 3: Scripted ACL Configuration	Elevate to security_admin, script Read ACL logic, and attach CRUD ACLs.
2:15 - 3:00	Milestone 4: Testing & Validation	Test record isolation via impersonation, validate CRUD access, and finalize documentation.

3. Resource Allocation & Pre-requisites

- Platform Environment: ServiceNow Personal Developer Instance (PDI).
- User Roles & Administrative Access:
 - System Administrator access (admin).
 - Security Administrator elevated privilege (security_admin).
- Scope Tracking: Update Set named ACL Project [Global] to capture all table schema, ACL scripts, and role configurations.

4. Risk & Mitigation Strategy

- Risk 1: Locking out admin users from records during ACL scripting.
 - *Mitigation:* Explicitly declare if (gs.hasRole('admin')) return true; at the top of the script before evaluating custom user conditions.
- Risk 2: Unintended inheritance of default table access.
 - *Mitigation:* Ensure Extends for is set to false during custom table creation to avoid inheriting conflicting ACL rules from parent tables.
- Risk 3: Incomplete ACL evaluation due to missing role elevation.
 - *Mitigation:* Always elevate to security_admin prior to navigating to System Security \rightarrow Access Control (ACL).